

Research paper

Meaningful Change in Depression Symptoms Assessed with the Patient Health Questionnaire (PHQ-9) and Montgomery-Åsberg Depression Rating Scale (MADRS) Among Patients with Treatment Resistant Depression in Two, Randomized, Double-blind, Active-controlled Trials of Esketamine Nasal Spray Combined With a New Oral Antidepressant

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ABSTRACT

Background: Patients with major depressive disorder who do not respond to ≥ 2 different pharmacological treatments within the current depressive episode are considered to have treatment resistant depression (TRD). This analysis determined meaningful change thresholds (MCT) of the Patient Health Questionnaire (PHQ-9) and Montgomery-Åsberg Depression Rating Scale (MADRS) using anchor-based methods and compared proportions of meaningful changes in patients with TRD across treatment groups from two Phase 3 trials for esketamine nasal spray (SPRAVATOTM).

Methods: Data from two Phase 3 trials in patients with TRD, TRANSFORM-1 and -2, were used in this analysis. The MCTs for the PHQ-9 and MADRS were derived using a clinician global impression of severity anchor. Blinded probability density functions displayed score distributions between anchor categories. Proportions of meaningful response were compared between treatment groups using chi-square tests supported by unblinded cumulative distribution functions of change scores.

Results: Baseline scores were similar for the PHQ-9 and MADRS between the esketamine/antidepressant (AD) and AD/placebo groups. The most appropriate MCT on the PHQ-9 was -6 points. By Day 28, 86.5% of patients reached or exceeded the PHQ-9 MCT in the esketamine/AD group compared to 70% in the placebo/AD group. The most appropriate MCT for the MADRS was -10 points. By Day 28, 78.2% of patients reached or exceeded the MADRS MCT in the esketamine/AD group compared to 65.0% in the placebo/AD group.

Conclusions: Individual-level meaningful change for the PHQ-9 and MADRS was effectively quantified using a clinical anchor to interpret efficacy from patients with TRD and their treating clinicians.

Abbreviations: CDF, cumulative distribution function; CGI-S, Clinical Global Impression – Severity; CI, confidence interval; FAP, full analysis population; FAS, full analysis set; MADRS, Montgomery-Åsberg Depression Rating Scale; MCID, minimal clinically important difference; MCT, meaningful change threshold; MDD, major depressive disorder; PDF, probability density function; PHQ-9, Patient Health Questionnaire; PRO, patient reported outcome; SD, standard deviation; SEM, standard error of measurement; SES, standard effect size; TRD, treatment resistant depression.

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Introduction

Major depressive disorder (MDD) is associated with excess morbidity and mortality and is a leading cause of disability worldwide in terms of total years lost due to disability (Baldessarini et al., 2017; World Health Organization. Depression Fact Sheet, 2020). About one-third of patients with MDD fail to achieve remission with multiple biogenic amine-based (eg, serotonin, norepinephrine) antidepressant treatments and hence have treatment-resistant depression (TRD) (Fava, 2003). In patients who respond to biogenic amine antidepressants, they remain symptomatic and at risk of self-harm during the typical 3 to 7 weeks to onset of effect (Rush et al., 2006). There is an unmet need to develop treatments providing effective, rapid-acting, and sustained or long-term relief of depressive symptoms, especially in patients with TRD.

Esketamine, the S-enantiomer of ketamine racemate and an N-methyl-D-aspartate receptor antagonist, has been recently approved as a nasal spray for the treatment of TRD (Daly et al., 2018; Fedgchin et al., 2019; Popova et al., 2019; Ochs-Ross et al., 2020).

Patient reported outcome (PRO) measures are effective at capturing the impact of disease on patients' daily lives, including symptom burden, functional abilities, and overall health-related quality of life (FDA, 2009). As these describe patient experience, it is important to provide results framed in meaningful terms. Methods of interpreting patient reported outcomes in terms of the minimal clinically important difference (MCID) were introduced in the late 1980's (Jaeschke et al., 1989). Use of the term MCID was phased out a decade later in favor of the minimal important difference. The purpose of this shift was to remove the focus on what was "clinically important," as the differences in the health-related quality of life are intended to be important to the patient, not the clinician. A second paradigm shift occurred in the mid-2010's when Cappelleri and Bushmakin, 2014 described the important distinction between group-level meaningful difference and individual-level meaningful response. Average differences of change between treatment groups of PRO scores may be statistically significant, but statistical significance also depends on the magnitude of the difference, sample size, and distributional qualities of the change score. Alternatively, patients can be classified as responders based on their change from baseline achieving a meaningful threshold. Around the same time, the Critical Path Institute's PRO Consortium and the Consensus Panel for Outcomes Measurement and Psychometrics: Advancing the Scientific Standards (COMPASS) pushed for preferential use of the term meaningful change threshold (MCT), which describes the threshold of change at which the change becomes meaningful to the patient (Critical Path Institute PRO Consortium, 2016; Hudgens et al., 2016). As a result, the MCT is a more recently used terminology for quantifying patient-level meaningful change.

The preferential method in a clinical trial and regulatory setting for derivation of MCTs is via anchor-based methods estimated as the mean change in a PRO measure in patients who reported an amount of change consistent with a clinically relevant change in global impression of disease severity or health measure with supportive cumulative distribution and probability density function curves (Coon and Cook, 2018; FDA, 2018; FDA, 2019; Wyrwich et al., 2013). In comparison, distribution-based methods use characteristics of the sample, such as one half of the standard deviation, to define an MCT. Although both methods can be used to measure meaningful change, European and United States regulatory agencies prefer anchor-based methods over distribution-based methods and suggest distribution derived thresholds to be used as supportive (Coon and Cook, 2018). In addition to these methods, there are many researcher-made decisions that influence the derivation and choice of an MCT. In most cases, the MCT is derived through a multifaceted triangulation of several approaches that cannot be prescribed in an analysis plan (Revicki et al., 2008).

The primary efficacy evaluation in the two esketamine Phase 3 trials described below was the MADRS, a clinician reported instrument used to measure depression severity. Measurement of patient reported

depression symptoms using the 9-item Patient Health Questionnaire (PHQ-9) was also included in these studies to provide additional information to evaluate the efficacy of esketamine in treating TRD.

The objective of this analysis was to assess meaningful change and derive an MCT for the PHQ-9 and MADRS using anchor-based methods and apply the derived MCT to compare meaningful differences in patients in response to the different treatment groups.

Methods

Study design

TRANSFORM-1 and TRANSFORM-2 are two Phase 3, randomized, double-blind, multicenter, active-controlled studies to evaluate the efficacy, safety and tolerability of fixed and flexible doses of esketamine nasal spray (56 mg or 84 mg), respectively, in combination with an oral antidepressant.

In the TRANSFORM-1 study, 346 patients were randomized 1:1:1 to receive double-blind nasal spray treatment with either one of 2 fixed doses of esketamine nasal spray (56 mg or 84 mg) or placebo (Fedgchin et al., 2019). In the TRANSFORM-2 study, 227 patients were randomized 1:1 to receive double-blind nasal spray treatment with flexible dose esketamine (56 mg and 84 mg) or placebo (Popova et al., 2019). In both studies, all patients initiated a new, open-label, oral antidepressant (AD) (duloxetine, escitalopram, sertraline or venlafaxine extended release). The double-blind treatment phase was 4 weeks in duration.

Patients were between 18 and 64 years of age with recurrent MDD (per Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition criteria) or single episode MDD (≥ 2 years), without psychotic features, confirmed by the Mini-International Neuropsychiatric Interview. Patients had moderate to severe depression (Inventory of Depressive Symptomatology total score ≥ 34 and MADRS total score ≥ 28). Patients had received separate therapeutic trials of at least 2 different antidepressants with nonresponse ($\leq 25\%$ improvement) in the current depressive episode at the time of randomization.

The study protocol was reviewed by an institutional review board. Adverse events were reported to the institutional review board per the sponsor's standard operating procedures.

Assessment instruments

9-item patient health questionnaire

This is a 9-item patient reported measure that assesses symptoms of depression and is used to measure the response to treatment (Kroenke et al., 2001). Each item is rated on a 4-point scale, and the responses are summed to provide a total score ranging from 0 to 27 with higher scores indicating a greater frequency of symptoms. The recall period is 2 weeks. Studies have shown the internal consistency and test-retest reliability of the PHQ-9 is excellent (Kroenke et al., 2001; Löwe et al., 2004). Additionally, in a trial setting, the PHQ-9 has been demonstrated to be a valid outcome measure able to accurately assess change in symptom frequency (Al-Harbi, 2012). However, to the best of our knowledge, an anchor-based MCT for the PHQ-9 in patients with TRD has not been established. The PHQ-9 was completed at baseline, Day 15, and Day 28.

Montgomery-asberg depression rating scale

This clinician-rated scale is designed to measure depression severity and detects changes due to antidepressant treatments. The scale consists of 10 items, each of which is scored from 0 (symptom not present or normal) to 6 (severe or continuous presence of the symptom), for a total possible score of 60. The MADRS evaluates apparent sadness, reported sadness, inner tension, sleep, appetite, concentration, lassitude, inability to feel (inner level), pessimistic thoughts, and suicidal thoughts. The test exhibits high inter-rater reliability. The typical recall period for the MADRS is 7 days. A higher score indicates a worsening of disease severity. The MADRS was conducted weekly from baseline to Day 28.

Clinical global impression – severity (CGI-S)

This clinician reported measure assesses the global severity of the illness based on the patient's history, psychosocial circumstances, symptoms, behavior, and the impact of the symptoms on the patient's ability to function (Spitzer et al., 1999). This instrument measures the severity of psychopathology in the patient from the clinician's perspective on a scale of zero to 7, with 7 being the most extremely ill patients. A global evaluation is conducted, where a higher score indicates a worsening case of the disease in terms of severity. The CGI-S was captured at various timepoints.

Statistical analysis

PRO analyses were conducted on the full analysis population (FAP) which included patients who received at least one dose of nasal spray study medication and one dose of oral antidepressant in the Double-blind Induction Phase who also had available PRO data on Day 1 (Baseline). For the longitudinal PRO analyses, the entrance criteria were expanded to include patients who had available PRO data at Day 1 (Baseline) and the respective post-baseline target assessment date (eg, Day 15, Day 28). Descriptive statistics were used to characterize patient demographics and clinical characteristics as well as instrument completion rates. All analyses were performed using SAS Version 9.4.

Meaningful change threshold (MCT)

The MCTs for the PHQ-9 and the MADRS were derived based on blinded data from the TRANSFORM-1 study using anchor-based methods and supported with distribution-based methods. The MCTs were then applied to the TRANSFORM-2 study data for the responder analysis.

For the anchor-based analysis, CGI-S was used as an anchor and patients were classified into response groups depending on their level of change in CGI-S over the course of the analysis. Mean (SE) within-patient change from baseline, along with min, max, 95% confidence intervals, and a p-value for the respective paired t-test were calculated for each of the categories of CGI-S change from baseline. In the categories of change, patients were classified among all possible change categories (15 levels, ranging from -7 to 7 where negative changes scores indicate improvement).

The standard effect size (SES) was calculated as the change score mean divided by the standard deviation of the change score (SES of Change). The SES and 95% CI was also calculated for the difference in mean change for each category compared to the adjacent anchor category. Additionally, the SES of Difference in Change was derived as the change score mean divided by the baseline standard deviation, which provided consideration of magnitude change from baseline PHQ-9 scores. Graphic representations of the anchor-based analysis were presented as probability density function (PDF) plots for change in PHQ-9 and MADRS total scores for each anchor group of change in CGI-S at each post-baseline visit using data from TRANSFORM-1. These PDFs allow for comparisons of the full distributions of PHQ-9 and MADRS change scores between CGI-S change anchor categories, which provide additional support for selecting the anchor category at which change in PHQ-9 and MADRS is observed.

The PDF plots used the Normal kernel function expressed as:

$$K_0(t) = \frac{1}{\sqrt{2\pi}} e^{-1/2t^2} \text{ for } -\infty < t < \infty$$

The smoothing approach used the default bandwidth free parameter selection method invoked by the UNIVARIATE procedure in SAS version 9 which minimizes approximate mean integrated standard error, also known as the expected L_2 risk function.

Distribution-based methods, which generate an absolute value for change, rather than directional, were used to supplement the anchor-based derivation of MCT. For the distribution-based MCT estimates, the mean patient data at baseline (Day 1) were used, calculated using $\frac{1}{2}$

of the pooled standard deviation (SD) at baseline and 1 standard error of measurement (SEM), defined as $SD \times [1 - \text{reliability}]^{1/2}$, where internal consistency reliability is Cronbach's alpha. Estimates of anchor-based and distribution-based MCTs from TRANSFORM-1 and TRANSFORM-2 were compared on a single graphic to assess the magnitude of meaningful change derived from each method and for comparison across the two trials.

Meaningful change from baseline in score

For both the PHQ-9 and the MADRS, the MCT was determined by evaluating the range of potential values from the categories of change anchor-based results and the distributional based results. Final determination was made through expert, clinical review.

Application of the MCT values to a trial population

The final MCTs for the PHQ-9 and MADRS were applied to data from TRANSFORM-2 in the subsequent unblinded responder analysis. Responder categories were created to classify patients from TRANSFORM-2 as "improved," "declined" or "stable," based on whether a change in the PHQ-9 or MADRS score met or exceeded the MCT. Predefined clinical categories of PHQ-9 and MADRS score were also defined and applied based on a review of the literature. For both the PHQ-9 and MADRS, meaningful change ("improved," "declined" or "stable") was defined as 1-group shift in predefined categories (PHQ-9 categories: Normal (0 to 4); Mild (5 to 9); Moderate (10 to 14); Moderately severe (15 to 19); Severe (≥ 20); MADRS categories: Normal/Symptom Absent (0-6); Mild (7-19); Moderate (20-34); Severe (>34)). Differences in responder status were evaluated using a χ^2 test.

To provide a more robust interpretive context for the responder analysis, cumulative distribution functions (CDFs) for those who improved ("responders") and those who remained stable or declined ("non-responders") were classified using the MADRS scores and plotted by treatment using data from TRANSFORM-2. The CDF analysis complements and extends understanding of responder rates by showing a complete picture of the proportions of patients exceeding thresholds along a continuum of change values between each treatment. The x-axis and y-axis are the change score of the measure and the raw frequency of responders for each change score, respectively.

Results

Baseline demographic and instrument measures (TRANSFORM-1 and TRANSFORM-2)

On average, patients in the FAP for TRANSFORM-2 were mostly middle-aged (mean = 46 years; SD = 11.9), non-Hispanic (93%), white (93%), and female (62%), with a mean of 12 years (SD = 10.2) since diagnosis with TRD (Table 1).

Patient demographics and clinical characteristics were generally similar in TRANSFORM-1 and across treatment groups (Table 1). Completion compliance of the PHQ-9 throughout the course of the analysis was high ($> 85\%$) at each visit from baseline until Day 28 (data not shown) and baseline PHQ-9 total scores were similar in the esketamine nasal spray (56 mg) + AD, esketamine nasal spray (84 mg) + AD, and AD + placebo nasal spray groups (TRANSFORM-1) and for the esketamine nasal spray (56 or 84 mg) + AD and AD + placebo nasal spray treatment groups (TRANSFORM-2).

MCT derivation: PHQ-9

In analyses using combined (blinded) treatment groups, the correlation between change on the CGI-S and change on the PHQ-9 at Day 28 was high (> 0.70) for the categories of change on the CGI-S anchor categories. Categories of change on the CGI-S were used to determine a

Table 1

Summary of Demographic and Clinical Characteristics (FAP; TRANSFORM-1 and TRANSFORM-2)

Characteristic	TRANSFORM-1				TRANSFORM-2		
	Esketamine Nasal Spray (56 mg) + Oral AD N = 115	Esketamine Nasal Spray (84 mg) + Oral AD N = 114	Placebo Nasal Spray + Oral AD N = 113	Total N = 342	Esketamine Nasal Spray (56/84 mg) + Oral AD N = 114	Placebo Nasal Spray + Oral AD N = 109	Total N = 223
Age (years)							
Mean (SD)	46.4 (11.18)	45.7 (11.10)	46.8 (11.36)	46.3 (11.19)	44.9 (12.58)	46.4 (11.14)	45.7 (11.89)
Median	48	47	47	47	45	47	47
Min, max	22, 64	18, 64	18, 64	18, 64	19, 64	20, 64	19, 64
Years since diagnosis							
Mean (SD)	16.1 (11.92)	13.6 (9.90)	15.0 (11.81)	14.9 (11.26)	12.9 (10.35)	11.1 (10.02)	12.0 (10.21)
Median	14	12	12	12	10.5	7	9
Min, max	0, 46	0, 44	0, 49	0, 49	0, 49	0, 50	0, 50
Baseline PHQ-9 Score							
Mean (SD)	20.3 (4.11)	20.7 (3.58)	20.8 (3.69)	20.6 (3.80)	20.2 (3.63)	20.4 (3.73)	20.3 (3.67)
Median	21	21	21	21	20.0	21.0	20.0
Min, max	7, 27	7, 27	12, 27	7, 27	5, 27	10, 27	5, 27
Baseline MADRS Score							
Mean (SD)	37.4 (4.76)	37.8 (5.58)	37.5 (6.16)	37.6 (5.51)	37.0 (5.69)	37.3 (5.66)	37.1 (5.67)
Median	37.0	37.5	37.0	37.0	37.0	37.0	37.0
Min, max	27, 50	25, 51	18, 53	18, 53	22, 48	21, 52	21, 52
Baseline CGI-S^a							
Mildly ill	0 (0.0%)	0 (0.0%)	1 (0.3%)	1 (0.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Moderately ill	18 (5.3)	24 (7.02)	14 (4.1)	56 (16.4%)	21 (18.6%)	19 (17.4%)	40 (17.9%)
Markedly ill	67 (19.6)	59 (17.3)	70 (20.5)	196 (57.3%)	64 (56.6%)	63 (57.8%)	127 (57.0%)
Severely ill	29 (8.5)	29 (8.5)	25 (7.3)	83 (24.3%)	27 (23.9%)	26 (23.9%)	53 (23.8%)
Among the most extremely ill patients	1 (0.3)	2 (0.6)	3 (0.9)	6 (1.8%)	1 (0.9%)	1 (0.9%)	2 (0.9%)
Sex, n (%)							
Male	34 (29.6%)	35 (30.7%)	32 (28.3%)	101 (29.53%)	39 (34.2%)	46 (42.2%)	85 (38.1%)
Female	81 (70.4%)	79 (69.3%)	81 (71.7%)	241 (70.47%)	75 (65.8%)	63 (57.8%)	138 (61.9%)
Race, n (%)							
White	91 (79.1%)	85 (74.6%)	86 (76.1%)	262 (76.61%)	106 (93.0%)	102 (93.6%)	208 (93.3%)
Black or African American	7 (6.1%)	7 (6.1%)	5 (4.4%)	19 (5.56%)	6 (5.3%)	5 (4.6%)	11 (4.9%)
Asian	2 (1.7%)	1 (0.9%)	2 (1.8%)	5 (1.46%)	1 (0.9%)	1 (0.9%)	2 (0.9%)
American Indian or Alaskan Native	0 (0.0%)	1 (0.9%)	0 (0.0%)	1 (0.29%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Multiple	0 (0.0%)	0 (0.0%)	1 (0.9%)	1 (0.29%)	1 (0.9%)	1 (0.9%)	2 (0.9%)
Other	8 (7.0%)	11 (9.6%)	10 (8.8%)	29 (8.48%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Not Reported	7 (6.1%)	9 (7.9%)	9 (8.0%)	25 (7.31%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Ethnicity, n (%)							
Not Hispanic or Latino	74 (64.3%)	78 (68.4%)	71 (62.8%)	223 (65.20%)	108 (94.7%)	99 (90.8%)	207 (92.8%)
Hispanic or Latino	33 (28.7%)	27 (23.7%)	31 (27.4%)	91 (26.61%)	5 (4.4%)	7 (6.4%)	12 (5.4%)
Not Reported	8 (7.0%)	8 (7.0%)	11 (9.7%)	27 (7.89%)	0 (0.0%)	1 (0.9%)	1 (0.5%)
Unknown	0 (0.0%)	1 (0.9%)	0 (0.0%)	1 (0.29%)	1 (0.9%)	2 (1.8%)	3 (1.3%)

Abbreviations: FAS = full analysis set; Max = maximum; Min = minimum; SD = standard deviation.^a Since depression severity (based on the IDS-C₃₀ and MADRS) was an inclusion criterion for entry in the study, the lowest 3 severity categories of the CGI-S were null and not included in the table.

threshold at which, on average, patients experience minimal meaningful improvement on the PHQ-9 at Day 28 (Table 2). Improvement on the PHQ-9 scale increased monotonically for all the categories from “no change (0)” to “1 category improvement (-1)” to “2 category improvement (-2)” to “3 category improvement (-3). The 95% CIs for mean

change on the PHQ-9 within the categories of change anchor categories did not overlap for the 0, -1, and -2 change groups providing evidence of unique groups. Specifically, a one category improvement (-1) is associated with a -6.7 point mean improvement in the PHQ-9 scale score at Day 28 compared to baseline, with a 95% CI [-8.22, -5.24; p<0.0001].

Table 2Anchor-Based Meaningful Change in PHQ-9 and MADRS from Baseline to Day 28 using the CGI-S^a (TRANSFORM-1)

Change in CGI-SAnchor Category	CGI-S Correlation	N	Mean (SD)	Min, Max	95% CI of Mean	P-Value ^b	SES of Change ^c	SES of Difference in Change ^c	95% CI of SES Difference
Change in PHQ-9									
Categories of change ^d	0.759								
-6		1	-23.0 (-)	-23, -23	-		-	-	-
-5		9	-21.1 (2.42)	-24, -18	(-22.97, -19.25)	< 0.0001	-8.72	-0.57	(-1.29, 0.16)
-4		45	-18.8 (4.30)	-24, -8	(-20.09, -17.51)	< 0.0001	-4.37	-0.69	(-1.11, -0.28)
-3		51	-15.6 (4.84)	-26, -4	(-16.97, -14.25)	< 0.0001	-3.22	-0.34	(-0.72, 0.04)
-2		58	-14.0 (4.54)	-23, -2	(-15.21, -12.82)	< 0.0001	-3.09	-1.23	(-1.6, -0.86)
-1		81	-6.7 (6.73)	-25, 6	(-8.22, -5.24)	< 0.0001	-1	-0.77	(-1.11, -0.43)
0		61	-2.2 (4.36)	-12, 6	(-3.35, -1.11)	0.0002	-0.51	-	-
1		10	-0.7 (4.22)	-11, 3	(-3.72, 2.32)	0.6124	-0.17	0.35	(-0.32, 1.02)
3		1	0.0 (-)	0, 0	-	-	-	-	-
Change in MADRS									
Categories of change ^d	0.771								
-5		8	-28.3 (7.52)	-39, -18	(-34.53, -21.97)	<.0001	-3.76	0.78	(0.01, 1.54)
-4		45	-34.6 (8.26)	-51, -5	(-37.10, -32.14)	<.0001	-4.19	-0.73	(-1.14, -0.31)
-3		52	-28.5 (8.60)	-49, -10	(-30.89, -26.11)	<.0001	-3.32	-0.82	(-1.2, -0.43)
-2		59	-21.3 (9.07)	-41, 3	(-23.62, -18.89)	<.0001	-2.34	-1.14	(-1.5, -0.78)
-1		80	-10.3 (10.02)	-37, 12	(-12.48, -8.02)	<.0001	-1.02	-0.83	(-1.17, -0.48)
0		61	-2.6 (7.96)	-29, 21	(-4.68, -0.60)	0.0120	-0.33	-	-
1		10	-0.9 (3.07)	-5, 4	(-3.10, 1.30)	0.3783	-0.29	0.23	(-0.44, 0.9)

Abbreviations: CGI-S = Clinician Global Impression of Severity; CI = confidence interval; MADRS = Montgomery-Åsberg Depression Rating Scale; Max = maximum; Min = minimum; PHQ-9 = 9-item Patient Health Questionnaire; SD = standard deviation; SES = standard effect size

^a The CGI-S evaluates the severity of psychopathology on a scale of 0 to 7, negative change from baseline indicates improvement. Considering total clinical experience, a patient is assessed on severity of mental illness at the time of rating according to: 0 = not assessed, 1 = normal, not at all ill, 2 = borderline mentally ill, 3 = mildly ill, 4 = moderately ill, 5 = markedly ill, 6 = severely ill, and 7 = among the most extremely ill patients.

^b P-value from paired t-test of mean difference from baseline = 0 with pooled standard deviation.

^c SES of Change is derived as mean change from baseline divided by the change score standard deviation. SES Difference in Change is derived as the difference in mean change from baseline for each category compared to the adjacent anchor category divided by the baseline standard deviation. Cohen's interpretability parameters for SES are 0.2 (Small), 0.5 (Moderate), and 0.8 (Large).

^d Categories were only displayed if sample size (N) was ≥ 1.

The difference between the “1 category improvement” and the “no change” groups had a large SES difference in change = -0.77; which is useful for interpreting the magnitude of this within-group change. The mean change in PHQ-9 at each category of change in CGI-S anchor is reported in Table 2. Distribution-based MCT thresholds of the baseline PHQ-9 score were 1.9 (1/2 SD) and 2.1 (SEM).

The blinded PDF plots of change in PHQ-9 stratified by CGI-S anchor change category from TRANSFORM-1 (Fig. 1) demonstrate separation of PDF curves between the CGI-S change categories. The change in PHQ-9 stratified by the categories of change on the CGI-S where substantially greater improvement on the PHQ-9 summary score was observed at 1 point of change on the CGI-S. All the categories of worsening or no change on the CGI-S are centered at 0 change in PHQ-9 score, while all improvement categories of the CGI-S are centered at values of improving on the PHQ-9.

Consideration of the MCT threshold for the PHQ-9 found the -6.7 point mean improvement associated with a 1-point improvement on the CGI-S, together with this body of evidence suggests an appropriate MCT of -6 points for the PHQ-9. The MCT for deterioration could not be derived from this study since the sample size for subjects experiencing deterioration was too small to provide a reliable estimate.

MCT derivation - MADRS

In analyses using combined (blinded) treatment groups, the correlation between change on the CGI-S and change on the MADRS at Day 28

were moderate to high (>0.60). Categories of change on the CGI-S were used to determine a threshold at which, on average, patients experience minimal meaningful improvement on the MADRS at Day 28 (Table 2). Improvement on the MADRS increased monotonically for all the categories from “no change (0)” to “1 category improvement (-1)” to “2 category improvement (-2)” to “3 category improvement (-3)” with the 95% CIs for the mean change non-overlapping for all groups with more than 10 patients. Further, a one category improvement (-1) is associated with a 10.3 point mean improvement in the MADRS score at Day 28 compared to baseline (95% CI [-12.48, -8.02]; $p < 0.0001$), and a large SES difference in change = -0.83 (95% CI [-1.17, -0.48]). Distribution-based MCT thresholds of the baseline MADRS score were 2.8 (1/2 SD) and 3.2 (SEM).

The blinded PDF plots of change in MADRS stratified by CGI-S anchor change category from study TRANSFORM-1 (Fig. 1) demonstrate separation of PDF curves between the CGI-S change categories. The change in MADRS stratified by the categories of change on the CGI-S where substantially greater improvement on the MADRS change score was observed at 1 point of improvement on the CGI-S and all other improvement categories. All the categories of worsening or no change on the CGI-S are centered at 0 change in MADRS score.

Consideration of the MCT threshold for the MADRS found a 10.3 point mean improvement associated with a 1-point improvement on the CGI-S, together with this body of evidence suggests an appropriate MCT of -10 points for the MADRS. The MCT for deterioration could not be derived from this study since the sample size for subjects experiencing

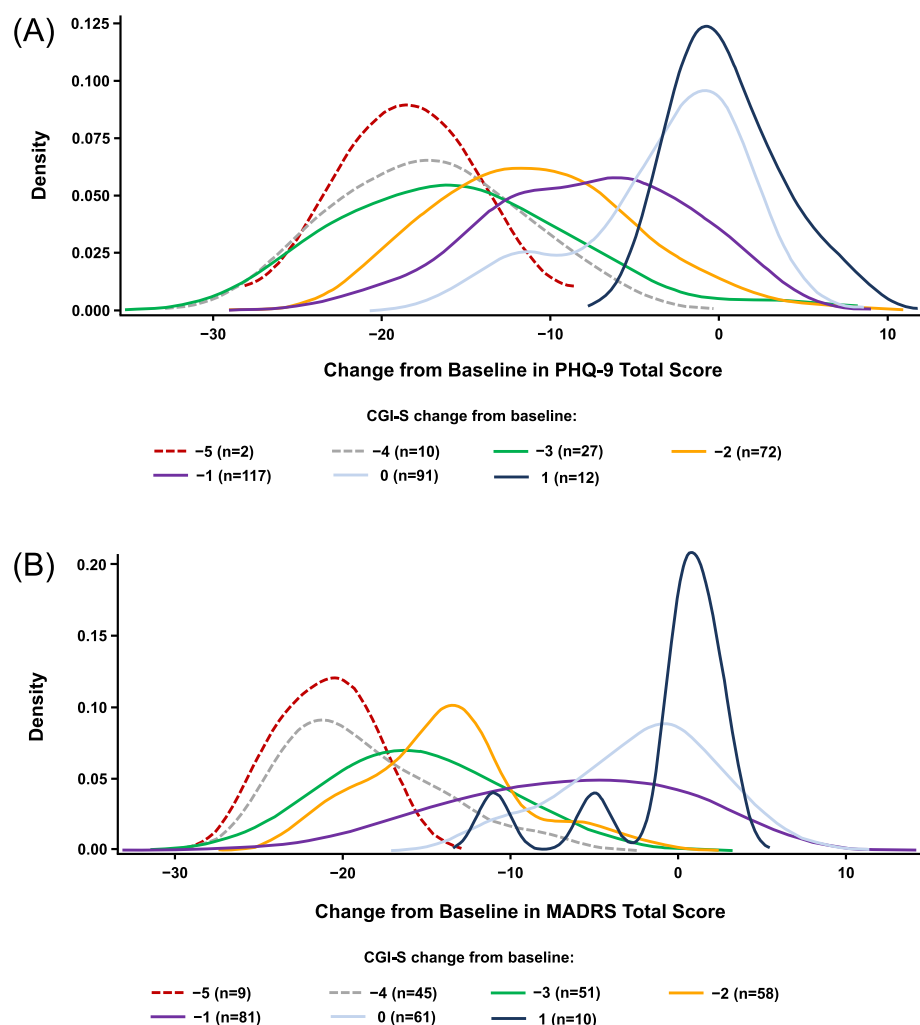


Fig. 1. Blinded Probability Density Function of the Change from Baseline in PHQ-9 and MADRS Total Score at Day 28 Stratified by Categories of Change on the CGI-S from TRANSFORM-1^a. (A) PHQ-9 Categories of Change on the CGI-S from Baseline. (B) MADRS Categories of Change on the CGI-S from Baseline.

Abbreviations: CGI-S = Clinician Global Impression of Severity; MADRS = Montgomery-Åsberg Depression Rating Scale; PHQ-9 = 9-item Patient Health Questionnaire

^a The probability density function curves show the distribution of change scores along the x-axis according to the category of anchor change. The y-axis displays the density at a particular point in change. The probability of observing a change score is the integral of the function over the range.

The probability density function plots of change in PHQ-9 and MADRS stratified by CGI-S anchor change category demonstrate separation of the curves between the CGI-S change categories.

deterioration was too small to provide a reliable estimate.

Application of the MCT values to a trial population

Meaningful changes in PHQ-9 from baseline (TRANSFORM-2)

Based on the selected MCT of -6 for the PHQ-9 (TRANSFORM-1), the distribution of response in the TRANSFORM-2 study (improved versus stable versus worsened) from baseline to Day 28 favored the esketamine/AD group over placebo/AD ($p = 0.0071$; Table 3). Specifically, the proportion of patients reaching or exceeding the MCT from baseline to Day 28 was 86.5% in the esketamine/AD group versus 70.0% in the placebo/AD group for a total difference of 16.5 percentage points in favor of esketamine/AD. Results for the predefined clinical categories had higher proportions of responders in each group (90.4% improvers in the esketamine/AD group versus 75.0% of improvers in the placebo/AD group) and were consistent for the between group difference (15.4%) with the results using the MCT. Unblinded CDF plots showed a separation between esketamine/AD group and placebo/AD at the derived MCT of -6 points of change as well as all change scores between -18 and 0 (less than 18-point improvement) for both Days 15 and 28 (Fig. 2).

Meaningful changes in MADRS from baseline (TRANSFORM-2)

Based on the selected MCT of -10 for the MADRS (TRANSFORM-1), the distribution of response in the TRANSFORM-2 study (improved versus stable versus worsened) from baseline to Day 28 favored the esketamine/AD group over placebo/AD ($p = 0.0162$; Table 3). Specifically, the proportion of patients reaching or exceeding the MCT from

baseline to Day 28 was 78.2% in the esketamine/AD group versus 65.0% in the placebo/AD group for a total difference of 13.2 percentage points in favor of esketamine/AD. Results for the predefined clinical categories of the MADRS had higher proportions of responders in each group (82.2% improvers in the esketamine/AD group versus 70.0% of improvers in the placebo/AD group) but were consistent for the between group difference (12.2%) with the results using the MCT. Unblinded CDF plots showed a separation between esketamine/AD and placebo/AD distributions at the derived MCT of -10 as well as all change scores between -25 and 0 for Day 15 and between -30 and 0 for Day 28 (Fig. 2).

Discussion

This is the first analysis to identify a meaningful change threshold on the PHQ-9 and the MADRS in patients with TRD using the described anchor-based MCT derivation. An MCT of -6 points was derived as the change in PHQ-9 associated with at least a 1-point improvement on the CGI-S from study TRANSFORM-1. After applying the MCT to data from TRANSFORM-2, the proportions of responders significantly favored esketamine/AD over placebo/AD. This value is supported by the fact that the mean changes are significantly different (Table 3) between “no change” and “1 category improvement” groups with separation of the respective 95% CIs for mean change for all categories. An MCT of -10 was selected and the change in MADRS score using the same methodology and rationale.

Table 3

Categorical Responder Analysis for Change from Baseline in PHQ-9 and MADRS Total Score by Treatment (TRANSFORM-2)

MCT Method	Visit	Response	Oral AD + Nasal Spray Placebo n (%)	Esketamine NasalSpray + Oral AD n (%)	P-value ^a
Change in PHQ-9					
Predefined Categories	Day 15	Improved	70 (67.3%)	84 (75.7%)	0.3913
		Stable	31 (29.8%)	25 (22.5%)	
		Worsened	3 (2.9%)	2 (1.8%)	
	Day 28	Improved	75 (75.0%)	94 (90.4%)	0.0074
		Stable	23 (23.0%)	9 (8.7%)	
		Worsened	2 (2.0%)	1 (1.0%)	
Anchor-based Derivation	Day 15	Improved	57 (54.8%)	74 (66.7%)	0.0932
		Stable	46 (44.2%)	37 (33.3%)	
		Worsened	1 (1.0%)	0 (0.0%)	
	Day 28	Improved	70 (70.0%)	90 (86.5%)	0.0071
		Stable	28 (28.0%)	14 (13.5%)	
		Worsened	2 (2.0%)	0 (0.0%)	
Change in MADRS					
Predefined Categories	Day 15	Improved	55 (53.9%)	74 (69.2%)	0.0748
		Stable	43 (42.2%)	31 (29.0%)	
		Worsened	4 (3.9%)	2 (1.9%)	
	Day 28	Improved	70 (70.0%)	83 (82.2%)	0.0496
		Stable	24 (24.0%)	17 (16.8%)	
		Worsened	6 (6.0%)	1 (1.0%)	
Anchor-based Derivation	Day 15	Improved	43 (42.2%)	58 (54.2%)	0.0484
		Stable	59 (57.8%)	47 (43.9%)	
		Worsened	0 (0.0%)	2 (1.9%)	
	Day 28	Improved	65 (65.0%)	79 (78.2%)	0.0162
		Stable	35 (35.0%)	20 (19.8%)	
		Worsened	0 (0.0%)	2 (2.0%)	

Abbreviations: MADRS = Montgomery-Åsberg Depression Rating Scale; MCT = Meaningful Change Threshold; PHQ-9 = 9-item Patient Health Questionnaire

^a Fisher's tests were used to test for differences in proportions of categorical change from baseline to Day 15 and Day 28 between treatment and placebo for each domain or summary score.

Primary evidence for MCT derivation of the PHQ-9

The MCT is intended to be the smallest observed change that is considered meaningful to the patient. The smallest categories of change on the CGI-S anchor following “no change” is the step to “minimal improvement,” which corresponds to improvement by 1-point on the CGI-S (Floden et al., 2019). For example, the primary evidence for selecting the PHQ-9 MCT was based on mean change for patients in the 1-point improvement category as well as the range of the upper and lower limit of the 95% confidence intervals (CIs). The observed lower bound of the CI for the 1-point improvement category could not overlap with the upper bound of the CI from the adjacent CGI-S “no change” category. Thus, ensuring the derived MCT was discriminative for patients experiencing actual change on the PHQ-9. It is important to note that this level of categorical change on the CGI-S corresponds with a significant and moderately large magnitude of change on the PHQ-9, as demonstrated by the -0.77 SES difference in change and significant p-value ($p < 0.05$).

Methods for MCT derivation

Evidence for an MCT of -6 point on the PHQ-9 was also supported by other, more qualitative and distribution-based approaches including clinician review. For example, results from the other anchor analyses provide additional insight on selection of the most appropriate MCT. Patients experiencing a 1-point change on the CGI-S score at Day 15 experienced an average change from baseline on the PHQ-9 of -7.9 (95% CI: -8.94, -6.78) points and rounding the lower bound of the 95% CI in the conservative direction would result in an MCT of -6 points (results not shown).

While the distribution-based methods (1/2 SD and SEM) have been

considered in legacy MCT methods, these parameters are sensitive to homogeneity in the distribution. More specifically, the patient population in this analysis were enrolled at the severe end of the PHQ-9 distribution with little variability resulting in MCTs less than 3 points to identify a patient as experiencing a meaningful change (approximately 11% change in the total score), which may be too small of a change to be considered meaningful.

Blinded PDFs of the PHQ-9 stratified by categories of change on the CGI-S demonstrated a clear and substantial shift in the distribution of PHQ-9 total score between no change (0) and a 1-point improvement on the CGI-S. The distributions of responses were divided by those who had no change or declined (positive change values) and those who improved. This consistent spacing of the PDF curves supports the use of CGI-S change categories for derivation of the MCT on the PHQ-9.

The MADRS anchor analysis results exhibited similar patterns thus the choice of MCT includes the same justification.

Application of derived MCT (Efficacy)

The proportion of patients with TRD who responded to esketamine/AD according to the patient-relevant MCT between baseline and Day 28 was significantly higher, 16.5 percentage points, compared to patients in the placebo/AD group ($p = 0.0071$). These results are supported by the separation between the unblinded CDFs of PHQ-9 change for esketamine/AD compared to placebo/AD group. The cumulative proportion of responders for most PHQ-9 values of improvement, e.g. between 0 and 16 points, suggests that esketamine/AD was favored over placebo/AD at a wide range of responder thresholds. Similar results on the MADRS CDFs support the difference in responder proportions across a range of MCT values. This is especially important given the understanding that a true definition of meaningful change is more likely a range of values rather than a single threshold.

A limitation of the analysis was that the observed treatment duration was short with only two post-baseline visits at which both the PHQ-9 and MADRS were assessed at the same visit over 28 days of treatment. Furthermore, while the PHQ-9 is a patient reported outcome that captures the patient's perspective, there was no available patient-reported global impression of change or severity to use as an anchor for MCT derivation. In addition, the CGI-S is a global measure based on clinician judgement assessing not only symptoms and behavior, but also the impact of the symptoms on the patient's ability to function. It would be important to confirm the proposed MCTs in future studies where a patient anchor may be available. However, the CGI-S still allowed for an anchor-based assessment using a simple and easily interpreted measure of severity. Another potential limitation of this work was the exclusive focus on differences in responder status and not in remission status.

Despite these limitations, this analysis is the first to offer a measure of meaningful change from the patient's perspective in an especially vulnerable treatment-resistant subpopulation of patients with MDD. The MADRS, used as the primary efficacy evaluation in these two trials, is a well-accepted scale to measure treatment-related differences in improvement of MDD symptoms. However, the MADRS is a clinician-administered measure, whereas the PHQ-9 is completed by the patient which is especially important in MDD where it can be difficult to objectively assess symptom severity and improvement in symptoms from the patient's perspective. The PHQ-9 has largely been used as a clinic-based diagnostic tool for MDD, with less frequent use in clinical trials as a measure of change or improvement in patient-reported symptoms of MDD. Furthermore, the PHQ-9 measures the frequency of the same DSM-5 symptoms for which the MADRS rates severity.

Conclusion

The current analysis is the first to derive an MCT, proposed to be -6, on the PHQ-9 to measure meaningful change from the perspective of the patient using regulatory preferred psychometric anchor-based

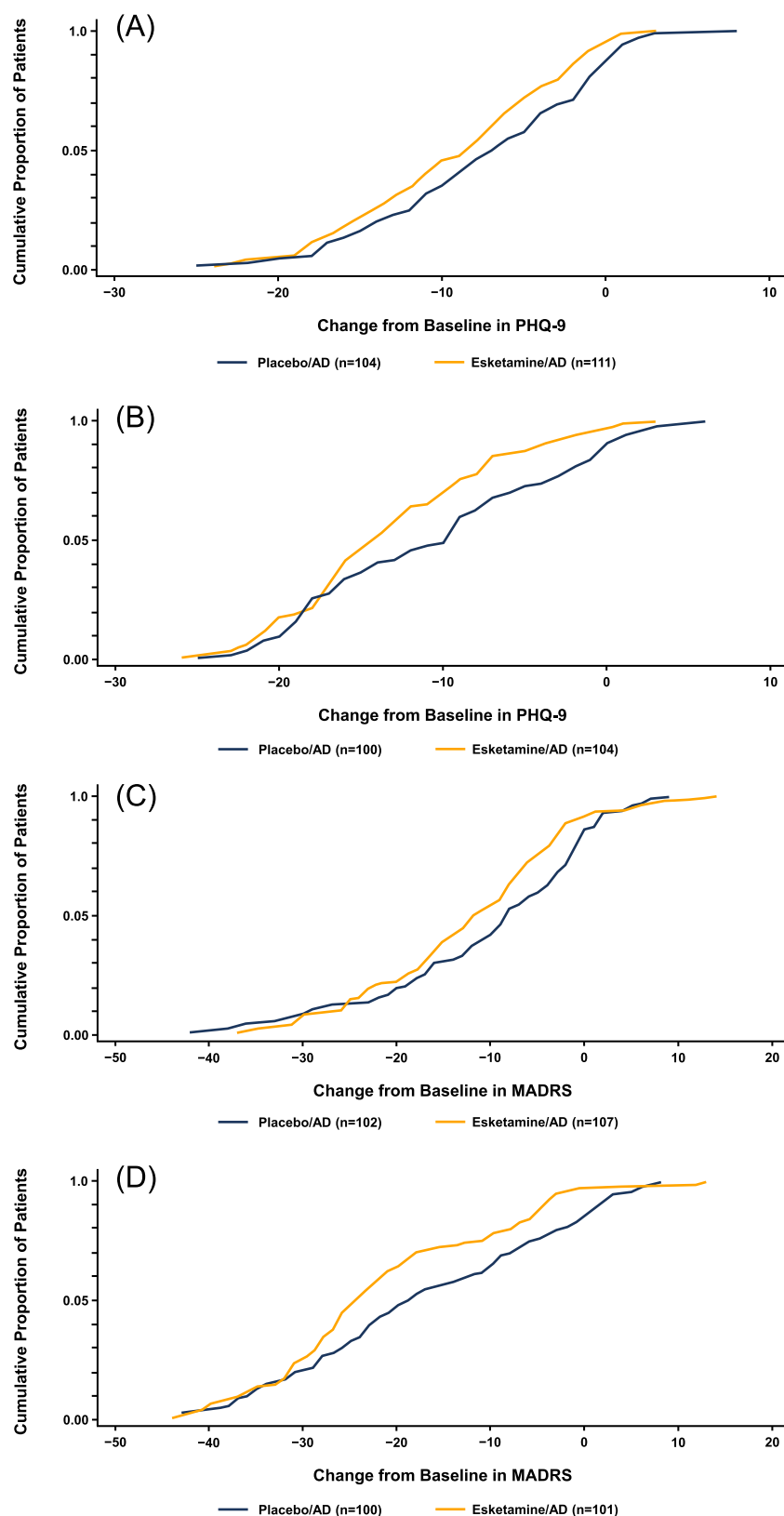


Fig. 2. Unblinded Cumulative Distribution Function Plots for Change from Baseline to Days 15 and 28 on the PHQ-9 and MADRS by Treatment Group (TRANSFORM-2). (A) PHQ-9 Day 15. (B) PHQ-9 Day 28. (C) MADRS Day 15. (D) MADRS Day 28. Abbreviations: CDF = Cumulative Distribution Function; CGI-S = Clinician Global Impression of Severity; MADRS = Montgomery-Åsberg Depression Rating Scale; PHQ-9 = 9-item Patient Health Questionnaire.

CDF plots of meaningful change in PHQ-9 show a separation between esketamine/AD group and placebo/AD at the derived MCT of -6 points of change as well as all change scores between -18 and 0 (less than 18-point improvement) for both Days 15 and 28. CDF plots of the meaningful change in MADRS show a separation between esketamine/AD and placebo/AD distributions at the derived MCT of -10 as well as all change scores between -25 and 0 for Day 15 and between -30 and 0 for Day 28.

methodology. This analysis provides substantial evidence that the PHQ-9, in addition to its utility as a diagnostic tool, can effectively measure change in depression symptoms among patients with TRD. This analysis also proposed the MCT to be -10 on the MADRS. The PHQ-9 should be considered as a key endpoint, alongside the MADRS, in clinical trials of treatment for MDD and TRD to fully characterize symptoms of depression from both a clinician and patient perspective. Furthermore, this analysis supports the positive benefits of esketamine/AD observed in the clinical program from the patient's perspective.

Contributors

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Declaration of Competing Interest

CJ, VP, MF, WD, KC, RL, and JS are employees of Janssen Pharmaceutical Companies of Johnson & Johnson. SH, LF, and MB declare no competing interests.

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References

- Al-Harbi, K.S., 2012. Treatment-resistant depression: therapeutic trends, challenges, and future directions. *Patient Prefer Adherence* 6, 369–388.
- Baldessarini, R., Forte, A., Selle, V., Sim, K., Tondo, L., Undurraga, J., Vazquez, G.H., 2017. Morbidity in depressive disorders. *Psychother Psychosom* 86, 65–72.
- Coon, C.D., Cook, K.F., 2018. Moving from significance to real-world meaning: methods for interpreting change in clinical outcome assessment scores. *Qual Life Res* 27 (1), 33–40.
- Critical Path Institute, 2016. PRO Consortium Webinar: Methods for determining clinically meaningful change Available at: <https://c-path.org/methods-for-determining-clinically-meaningful-change/>. Accessed October 7, 2020.
- Daly, E.J., Singh, J.B., Fedgchin, M., Cooper, K., Lim, P., Shelton, R.C., Thase, M.E., Winokur, A., Van Nueten, L., Manjil, H., 2018. Efficacy and safety of intranasal esketamine adjunctive to oral antidepressant therapy in treatment-resistant depression: a randomized clinical trial. *JAMA Psychiatry* 75(2), 139–148.
- Fava, M., 2003. Diagnosis and definition of treatment-resistant depression. *Biol Psychiatry* 53, 649–659.
- Fedgchin, M., Trivedi, M., Daly, E., Melkote, R., Lane, R., Lim, P., Vitagliano, D., Blier, P., 2019. Efficacy and safety of fixed-dose esketamine nasal spray combined with a new oral antidepressant in treatment-resistant depression: results of a randomized, double-blind, active-controlled study (TRANSFORM-1). *Int J Neuropsychopharmacol* 22 (10), 616–630.
- Floden, L., Hudgens, S., Jamieson, C., et al., 2019. Evaluation of depression severity items on the Patient Health Questionnaire (PHQ-9) and Sheehan Disability Scale (SDS) in patients with treatment-resistant depression treated with esketamine nasal spray [poster]. In: Presented at the 32nd Annual Psych Congress. San Diego, CA, US. October 3–6, 2019.
- Food and Drug Administration, 2009. Guidance for industry: patient-reported outcome measures: use in medical product development to support labeling claims. *Fed Regist* 74 (235), 65132–65133.
- Food and Drug Administration, 2018. Patient-Focused Drug Development Guidance: Methods to Identify What is Important to Patients and Select, Develop or Modify Fit-for-Purpose Clinical Outcome Assessments. Available at: <https://www.fda.gov/drugs/news-events-human-drugs/patient-focused-drug-development-guidance-methods-identify-what-important-patients-and-select>. Accessed July 31, 2020.
- Food and Drug Administration, 2019. Public Workshop on Patient-Focused Drug Development: Guidance 4 – Incorporating Clinical Outcome Assessments into Endpoints for Regulatory Decision Making. Available at: <https://www.fda.gov/drugs/development-approval-process-drugs/public-workshop-patient-focused-drug-development-guidance-4-incorporating-clinical-outcome>. Accessed October 8, 2020.
- Hudgens, S., Symonds, T., McLeod, L., Coon, C., 2016. Moving the science forward: psychometric considerations and study designs for understanding meaningful change and conducting mixed methods research. In: Presented at the 2016 ISPOR 19th Annual European Congress. Vienna, Austria. October 31, 2016.
- Jaeschke, R., Singer, J., Guyatt, G.H., 1989. Measurement of health status: ascertaining the minimal clinically important difference. *Control Clin Trials* 10 (4), 407–415.
- Kroenke, K., Spitzer, R.L., Williams, J.B., 2001. The PHQ-9: validity of a brief depression severity measure. *J Gen Intern Med* 16 (9), 606–613.
- Löwe, B., Kroenke, K., Herzog, W., Gräfe, K., 2004. Measuring depression outcome with a brief self-report instrument: sensitivity to change of the Patient Health Questionnaire (PHQ-9). *J Affect Disord* 81 (1), 61–66.
- Ochs-Ross, R., Daly, E., Zhang, Y., Lane, R., Lim, P., Morrison, R., Hough, D., Manji, H., Drevets, W., Sanacora, G., Steffens, D., Adler, C., McShene, R., Gaillard, R., Wilkinson, S., Singh, J., 2020. Efficacy and safety of esketamine nasal spray plus an oral antidepressant in elderly patients with treatment-resistant depression TRANSFORM-3. *Am J Geriatr Psychiatry* 28 (2), 121–141.
- Popova, V., Daly, E., Trivedi, M., Cooper, K., Lane, R., Lim, P., Mazzucco, C., Hough, D., Thase, M., Shelton, R., Molero, P., Vieta, E., Bajbouj, M., Manji, H., Drevets, W., Singh, J., 2019. Efficacy and safety of flexibly dosed esketamine nasal spray combined with a newly initiated oral antidepressant in treatment-resistant depression: a randomized double-blind active-controlled study. *Am J Psychiatry* 176 (6), 428–438.
- Revicki, D., Hays, R.D., Cella, D., Sloan, J., 2008. Recommended methods for determining responsiveness and minimally important differences for patient-reported outcomes. *J Clin Epidemiol* 61 (2), 102–109.
- Rush, A., Trivedi, M., Wisniewski, S., Nierenberg, A., Stewart, J., Warden, D., Niederrehe, G., Thase, M., Lavori, P., Lebowitz, B., McGrath, P., Rosenbaum, J., Sackeim, H., Kupfer, D., Luther, J., Fava, M., 2006. Acute and longer-term outcomes in depressed outpatients requiring one or several treatment steps: a STAR*D report. *Am J Psychiatry* 163, 1905–1917.
- Spitzer, R.L., Kroenke, K., Williams, J.B., 1999. Validation and utility of a self-report version of PRIME-MD: the PHQ primary care study. *JAMA* 282 (18), 1737–1744.
- World Health Organization. Depression Fact Sheet, Updated March 2018. Available at: <http://www.who.int/en/newsroom/fact-sheets/detail/depression>. Accessed April 24, 2020.
- Wyrrich, K.W., Norquist, J.M., Lenderking, W.R., Acaster, S., Industry Advisory Committee of International Society for Quality of Life Research (ISOQOL), 2013. Methods for interpreting change over time in patient-reported outcome measures. *Qual Life Res* 22 (3), 475–483.